Let Φ_B be the magnetic flux linked with each turn of the coil of N turns, then the total flux linked with the coil $N\Phi_B$ (flux linkage) is proportional to the current i in the coil.

$$N\Phi_B \propto i$$

$$N\Phi_B = Li \qquad (4.15)$$

(or) $$L = \frac{N\Phi_B}{i}$$

The constant of proportionality L is called self-inductance of the coil. It is also referred to as the coefficient of self-induction. If $i = 1A$, then $L = N\Phi_B$. **Self-inductance or simply inductance of a coil is defined as the flux linkage of the coil when 1A current flows through it.**

When the current i changes with time, an emf is induced in it. From Faraday's law of electromagnetic induction, this self-induced emf is given by

$$\varepsilon = -\frac{d(N\Phi_B)}{dt} = -\frac{d(Li)}{dt}$$

(using equation 4.15)

$$\therefore \varepsilon = -L\frac{di}{dt} \qquad (4.16)$$

(or) $$L = \frac{-\varepsilon}{di/dt}$$

The negative sign in the above equation means that the self-induced emf always opposes the change in current with respect to time. If $di/dt = 1\,A\,s^{-1}$, then $L = -\varepsilon$. **Inductance of a coil is also defined as the opposing emf induced in the coil when the rate of change of current through the coil is 1 $A\,s^{-1}$.**

Unit of inductance

Inductance is a scalar and its unit is $Wb\,A^{-1}$ or $V\,s\,A^{-1}$. It is also measured in henry (H).

$1\,H = 1\,Wb\,A^{-1} = 1\,V\,s\,A^{-1}$

The dimensional formula of inductance is $M\,L^2\,T^{-2}A^{-2}$.

If i = 1 A and $N\Phi_B = 1\,Wb\,turns$, then $L = 1H$.

Therefore, **the inductance of the coil is said to be one henry if a current of 1 A produces unit flux linkage in the coil.**

If $di/dt = 1\,A\,s^{-1}$ and $\varepsilon = -1\,V$, then $L = 1H$.

Therefore, **the inductance of the coil is one henry if a current changing at the rate of 1 $A\,s^{-1}$ induces an opposing emf of 1 V in it.**

Physical significance of inductance

We have learnt about inertia in XI standard. In translational motion, mass is a measure of inertia; in the same way, for rotational motion, moment of inertia is a measure of rotational inertia (Refer sections 3.2.1 and 5.4 of XI physics text book). Generally, inertia means opposition to change its state.

The inductance plays the same role in a circuit as mass and moment of inertia play in mechanical motion. When a circuit is switched on, the increasing current induces an emf which opposes the growth of current in a circuit (Figure 4.20(a)). Likewise, when circuit is broken, the decreasing current induces an emf in the reverse direction. This emf now opposes the decay of current (Figure 4.20(b)).

Thus, inductance of the coil opposes any change in current and tries to maintain the original state.

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